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Find the Maclaurin series for the given function.

$$\frac{10}{1+x}$$

Let f be a function with derivative of all orders throughout some interval containing 0 as an interior point. Then the Maclaurin series of f is the Taylor series generated by f at $x = 0$.

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots + \frac{f^{(n)}(0)}{n!}x^n + \dots$$

First find $f(0)$.

$$f(x) = \frac{10}{1+x}$$

$$\begin{aligned} f(0) &= \frac{10}{1+(0)} \\ &= 10 \end{aligned}$$

Find a general formula for $f^{(n)}(x)$. Start by finding $f'(x)$. Differentiate $f(x)$ with respect to x .

$$f(x) = \frac{10}{1+x}$$

$$f'(x) = -\frac{10}{(1+x)^2}$$

Now find $f'(0)$.

$$f'(x) = -\frac{10}{(1+x)^2}$$

$$\begin{aligned} f'(0) &= -\frac{10}{(1+0)^2} \\ &= -10 \end{aligned}$$

Now find $f''(x)$. Differentiate $f'(x)$ with respect to x .

$$f'(x) = -\frac{10}{(1+x)^2}$$

$$f''(x) = \frac{20}{(1+x)^3}$$

Find $f''(0)$.

$$f''(x) = \frac{20}{(1+x)^3}$$

$$\begin{aligned} f''(0) &= \frac{20}{(1+0)^3} \\ &= 20 \end{aligned}$$

Find $f'''(x)$. Differentiate $f''(x)$ with respect to x .

$$f''(x) = \frac{20}{(1+x)^3}$$

$$f'''(x) = -\frac{60}{(1+x)^4}$$

Find $f'''(0)$.

$$f'''(x) = -\frac{60}{(1+x)^4}$$

$$\begin{aligned} f'''(0) &= -\frac{60}{(1+0)^4} \\ &= -60 \end{aligned}$$

Therefore, $f(0) = 10$, $f'(0) = -10$, $f''(0) = 20$, $f'''(0) = -60$.

So $f(x) = \frac{10}{1+x}$, $f'(x) = -\frac{10}{(1+x)^2}$, $f''(x) = \frac{20}{(1+x)^3}$, and $f'''(x) = -\frac{60}{(1+x)^4}$. Use these results to find a general formula for $f^{(n)}(x)$.

$$f^{(n)}(x) = \frac{10(-1)^n(n!)}{(1+x)^{n+1}}$$

Next, find a general formula for $f^{(n)}(0)$. This can either be done by identifying a pattern in the values of $f(0)$, $f'(0)$, $f''(0)$, and $f'''(0)$ or by substituting 0 for x in the general formula for $f^{(n)}(x)$. Here, we will substitute 0 for x in the formula for $f^{(n)}(x)$.

$$f^{(n)}(x) = \frac{10(-1)^n(n!)}{(1+x)^{n+1}}$$

$$\begin{aligned} f^{(n)}(0) &= \frac{10(-1)^n(n!)}{(1+0)^{n+1}} \\ &= 10(-1)^n(n!) \end{aligned}$$

Recall that the Maclaurin series of f was given by $\sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k$. Use the results of the previous step to write the Maclaurin series for

$$f(x) = \frac{10}{1+x}.$$

$$\sum_{k=0}^{\infty} 10(-1)^k(x)^k$$

Therefore, $\sum_{k=0}^{\infty} 10(-1)^k(x)^k$ is the Maclaurin series for $f(x) = \frac{10}{1+x}$.